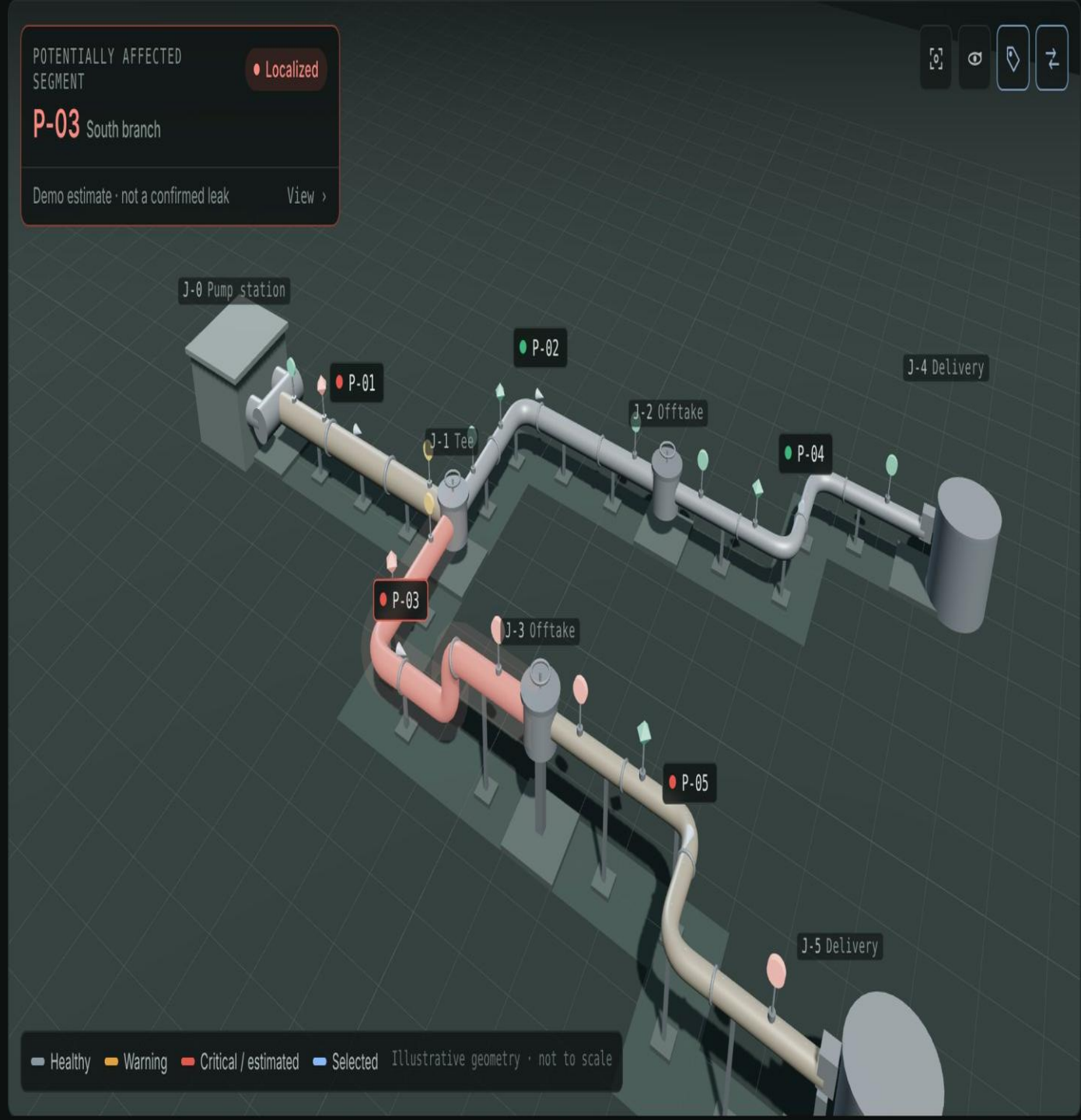
RESEARCH PROTOTYPE

SPS

Smart Piping System

Monitoring pipe-network sensors to flag abnormal behavior early and point operators to the segment that may be affected.

Pressure · Flow · Water level



Right: 3D view from the SPS interface · illustrative geometry, demo data

THE PROBLEM

Leaks waste resources quietly, and are often discovered late

Lost resource

Treated water, and the energy used to pump it, escape before anyone notices.

Growing damage

A small, unnoticed leak can grow into service interruptions and damage around the pipe.

Late, manual discovery

Releases are frequently reported by people on site rather than flagged by monitoring systems.

~17%

of hazardous-liquid pipeline releases were first identified by the pipeline control room.

The rest were first noticed by the public, field staff, emergency responders or patrols.

Source: Kiefner & Associates for U.S. DOT PHMSA, Leak Detection Study, Final Report 12-173 (2012); U.S. incidents Jan 2010 – Jul 2012.

Water losses are counted in billions of cubic metres a year

126 bn m³

lost worldwide every year as non-revenue water, worth roughly US$39 billion

Liemberger & Wyatt, reported by IWA The Source, 2018

32+ bn m³

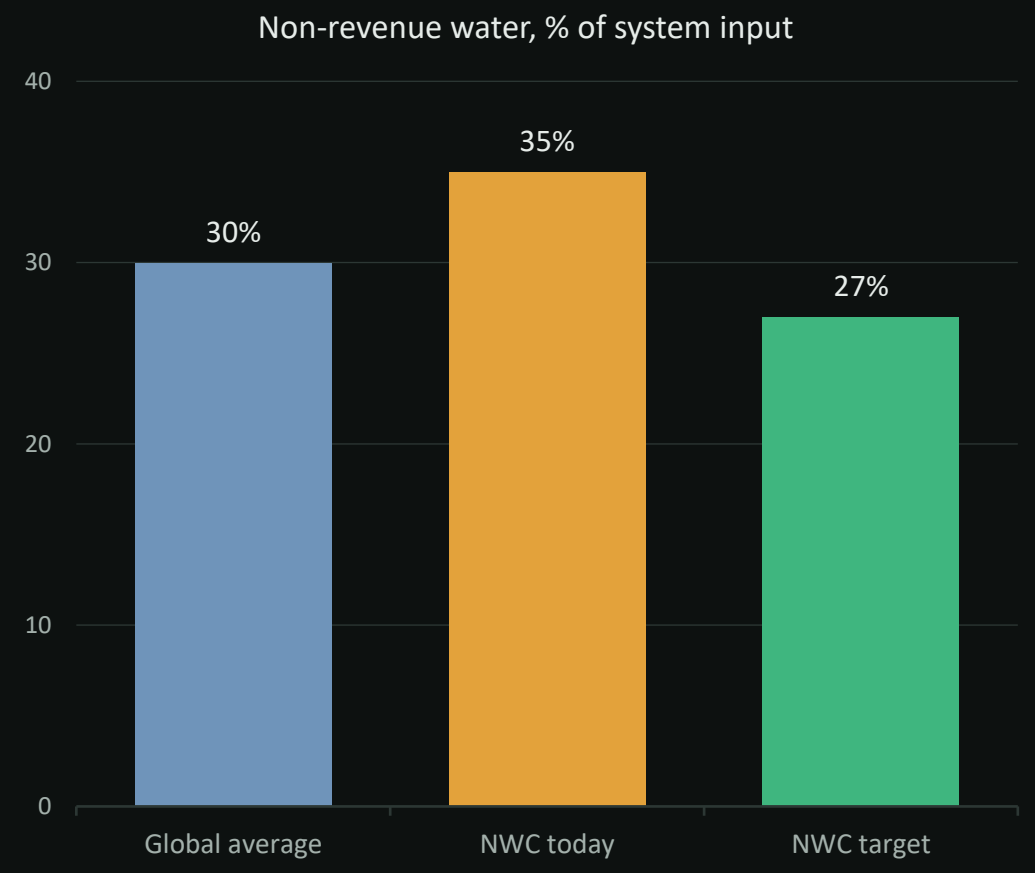
of treated water physically leak from urban supply systems each year

Kingdom, Liemberger & Marin, World Bank, 2006

35%

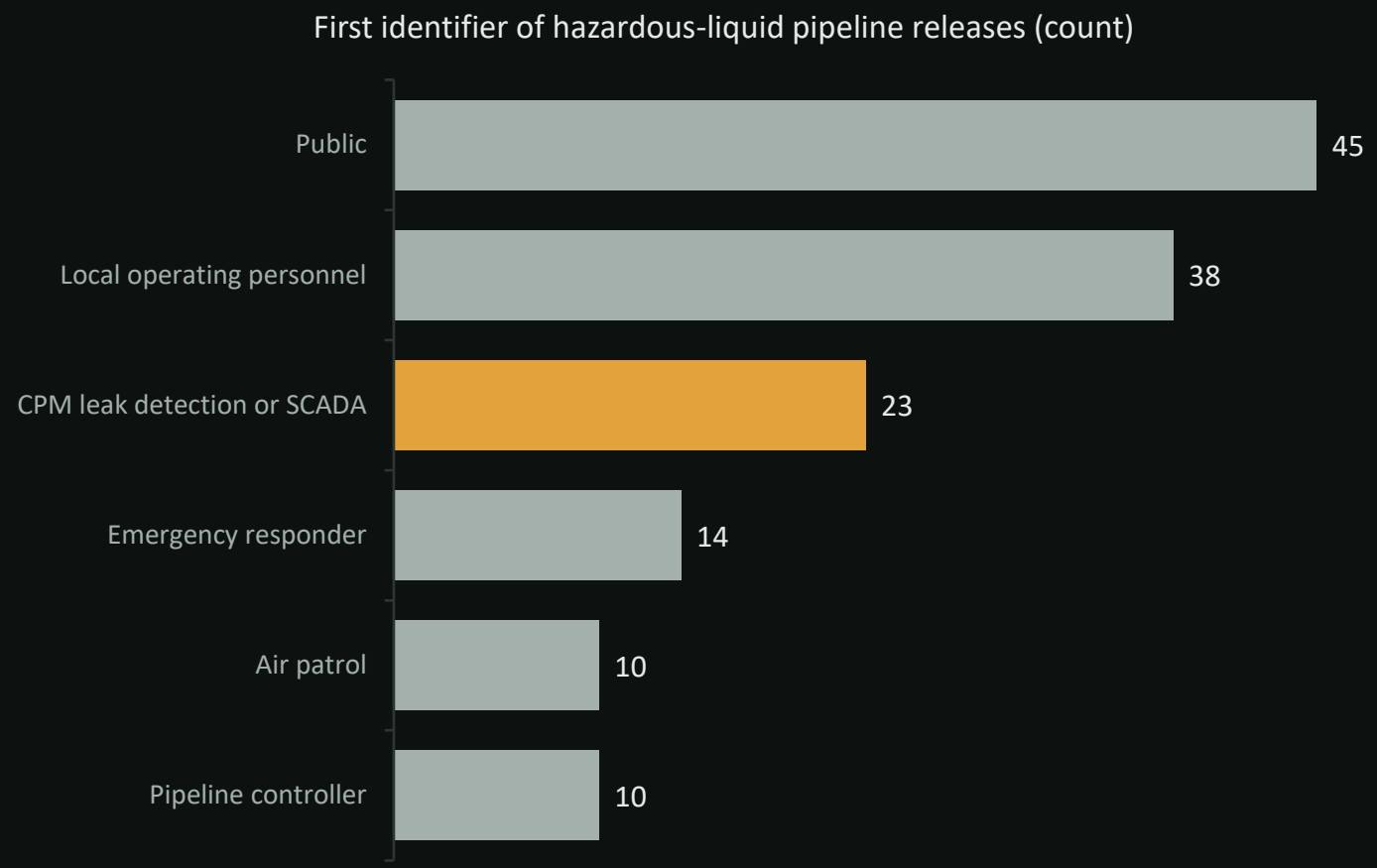
non-revenue water at Saudi Arabia's National Water Company, across a 134.7k km water network

NWC Overview, figures through March 2026



Global average ≈ 30% (Liemberger & Wyatt via IWA, 2018). NWC current 35% and target 27% from the NWC Overview (2026); the target year is not stated. Full links in References.

Most releases were first spotted by people, not by systems



20% of releases identified by CPM leak detection, where it was functional (17 of 86)

28% of releases identified by SCADA, where it was functional (43 of 152)

The gap SPS targets: continuous, automated screening that raises attention before a leak is noticed on site.

Source: Kiefner & Associates for U.S. DOT PHMSA, Leak Detection Study (Dec 2012), U.S. hazardous-liquid pipeline releases Jan 2010 – Jul 2012. Not water-network data.

OUR SOLUTION

SPS screens sensor data and shows operators where to look

Monitor

Read pressure, flow and water-level signals and compare each with its expected normal value.

Detect

Flag abnormal behavior with deviation rules and a machine-learning event classifier.

Localize

Estimate the affected pipe segment, or say clearly when evidence is ambiguous or insufficient.

Alert

Raise one alert per event, then investigate it in a 3D control room with trends and evidence.

Status today: a working software prototype on recorded lab data and simulated sensors. No field hardware is connected yet.

HOW IT WORKS

From a sensor reading to an operator alert in six steps

Read
Pressure, flow and water-level readings arrive per time step

Validate
Reject missing, non-numeric or out-of-range values; compare with normal reference

Detect
Deviation scoring and the event classifier flag abnormal behavior

Localize
Flow residual + endpoint pressure drop scored per segment

Alert
One alert per continuous event; no repeats while it lasts

Investigate
3D network, trends and evidence in the dashboard

Localization outcomes

Localized
one segment clearly leads

Ambiguous
candidates too close to call

Insufficient evidence
no local flow evidence

No leak pattern
signal within normal noise

Water-level rule (separate from the model): above 200 = normal · below 200 = potential leak indication · exactly 200 = boundary, reviewed without a new alert.

SYSTEM ARCHITECTURE

Three layers: data, a Python backend, two interfaces

DATA SOURCES

WDSEventDB lab recordings
Clean + Leakage workbooks

Synthetic network simulator
5 segments · 11 sensors · seeded

Simulated water level
threshold 200 · own random stream

Field IoT sensors
planned · not connected

SPS BACKEND · PYTHON PACKAGE sps/

`simulated_stream`
replays recorded rows in batches

`SPSInference`
saved Random Forest · 15 features

`simulated_localization`
segment scoring + adapter

`water_level`
threshold rule · event de-duplication

INTERFACES

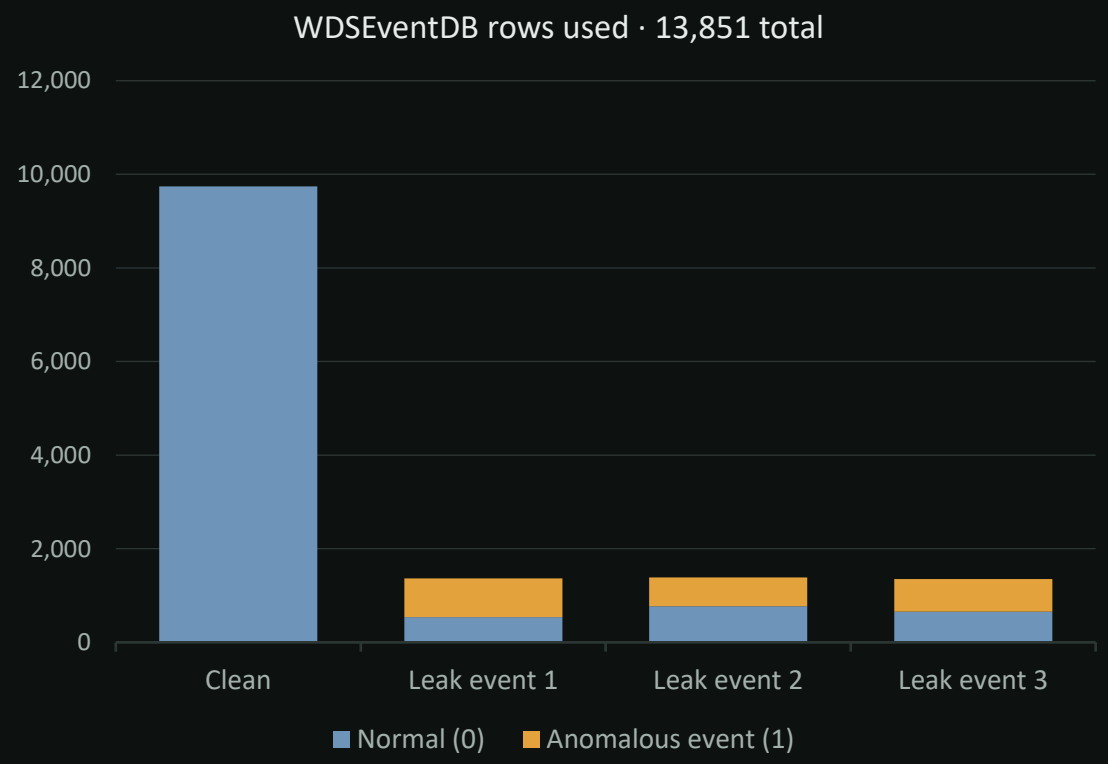
Streamlit dashboard
Plotly 3D network, trends, alerts

Web control room
Three.js + Chart.js · built-in demo data

SCADA / historian link
planned

Dashed = planned. The model never sees water level or localization inputs; each signal path stays separate.

Recorded lab events for learning, simulation for localization



Lab water-distribution testbed · 15 sensor features (4 pressure, 4 flow, 5 pump VFD, 2 valve) · no per-row timestamps

1. **Obtain** — WDSEventDB from its official Zenodo record; all 9 workbooks schema-checked
2. **Parse** — XLSX read with Python's standard library; no extra workbook packages
3. **Select** — Sensor readings only as inputs; Labels kept as the target, never as a feature
4. **Clean** — Median imputation; constant features dropped (VFD 2, VFD 4-1)
5. **Fit safely** — Preprocessing and model fitted on training rows or workbooks only
6. **Simulate** — Seeded synthetic network for localization and water-level scenarios

Label 1 = anomalous event sample in the source data, not necessarily a confirmed leak. Counts from reports/sps_supervised_baseline_results.txt.

TECHNOLOGY STACK

Built with a small, standard Python and web toolset

Python 3.10–3.12

Backend package, scripts and tests

NumPy

Signal maths, simulation, scoring

scikit-learn + joblib

Random Forest pipeline; saved model

Streamlit

Operations dashboard

Plotly

3D network and trend charts

Three.js + Chart.js

Web control room demo

unittest

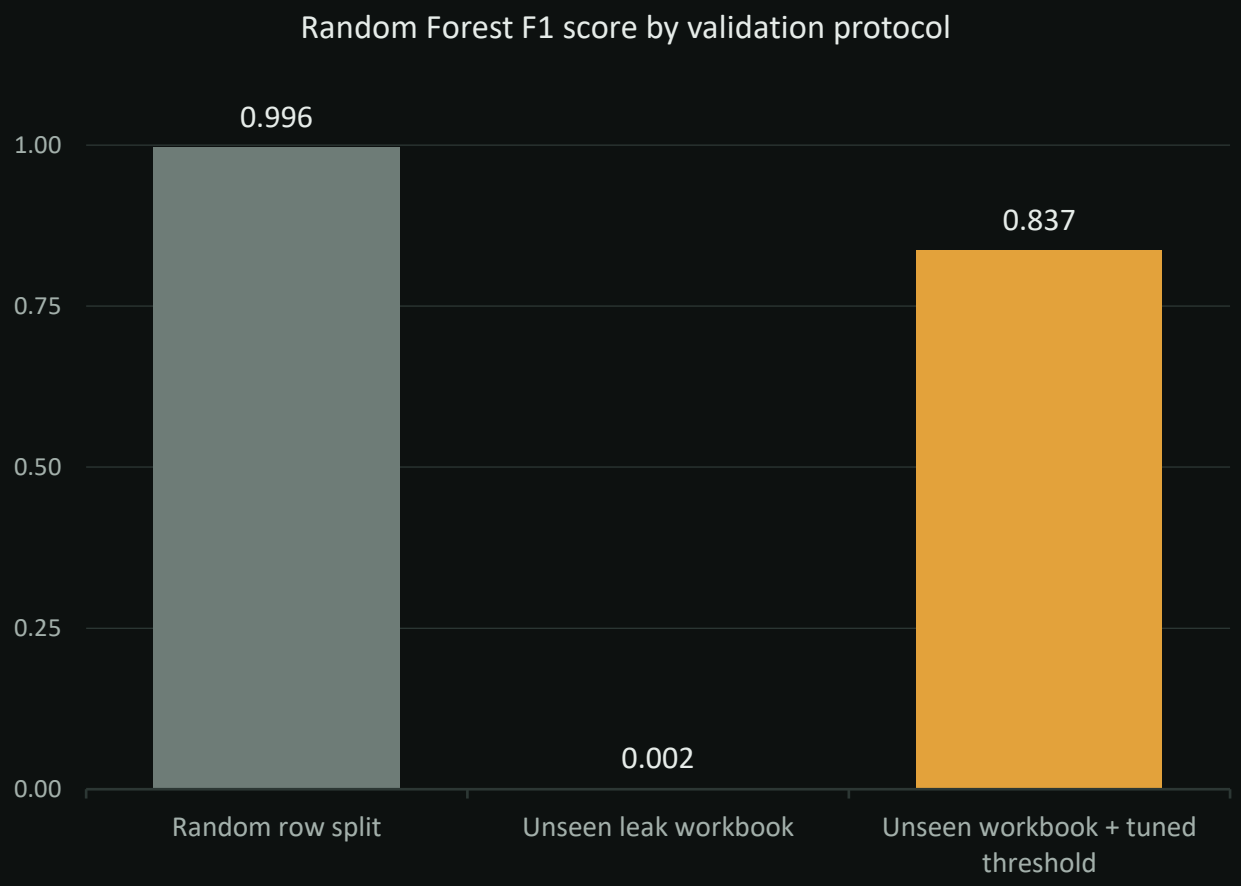
88 automated tests

Git

Version control, reproducible scripts

Hardware concept (planned, not yet integrated): Arduino controller with pressure, flow and HW-038 water-level sensors.

Random splits overstate how well the model generalizes



Random row split · F1 0.996

Rows from the same recording appear in training and test, so this number is optimistic.

Leave one leak workbook out · F1 0.002

On a recording it had never seen, the default model missed almost every anomalous sample.

Threshold tuned on a separate workbook · F1 0.837

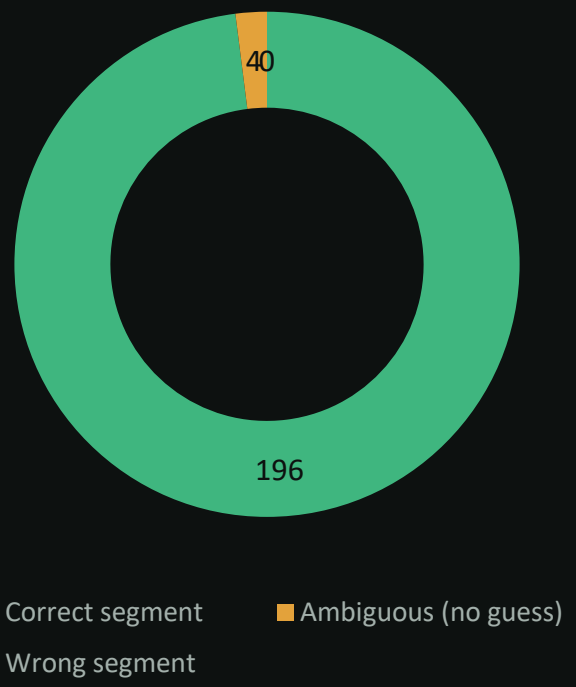
Recall 0.986, but 791 false alarms among 1,970 normal samples; thresholds varied by fold.

Takeaway: the classifier is a research baseline, not a field-ready detector. More varied, time-stamped events are needed.

Source: SPS reports sps_supervised_baseline_results, sps_grouped_evaluation, sps_threshold_calibration (WDSEventDB Clean + 3 Leakage workbooks).

Localization holds up in simulation and admits uncertainty

200 synthetic leak cases



5 segments × 4 leak strengths × 10 seeds, with varied sensor gain and noise. All 4 ambiguous cases were on P-04.

Edge cases

No event
→ **No leak pattern**

Weak event
→ **No leak pattern**

Pressure-only change
→ **Insufficient evidence**

Same flow rise everywhere
→ **Ambiguous**

Automated checks

88

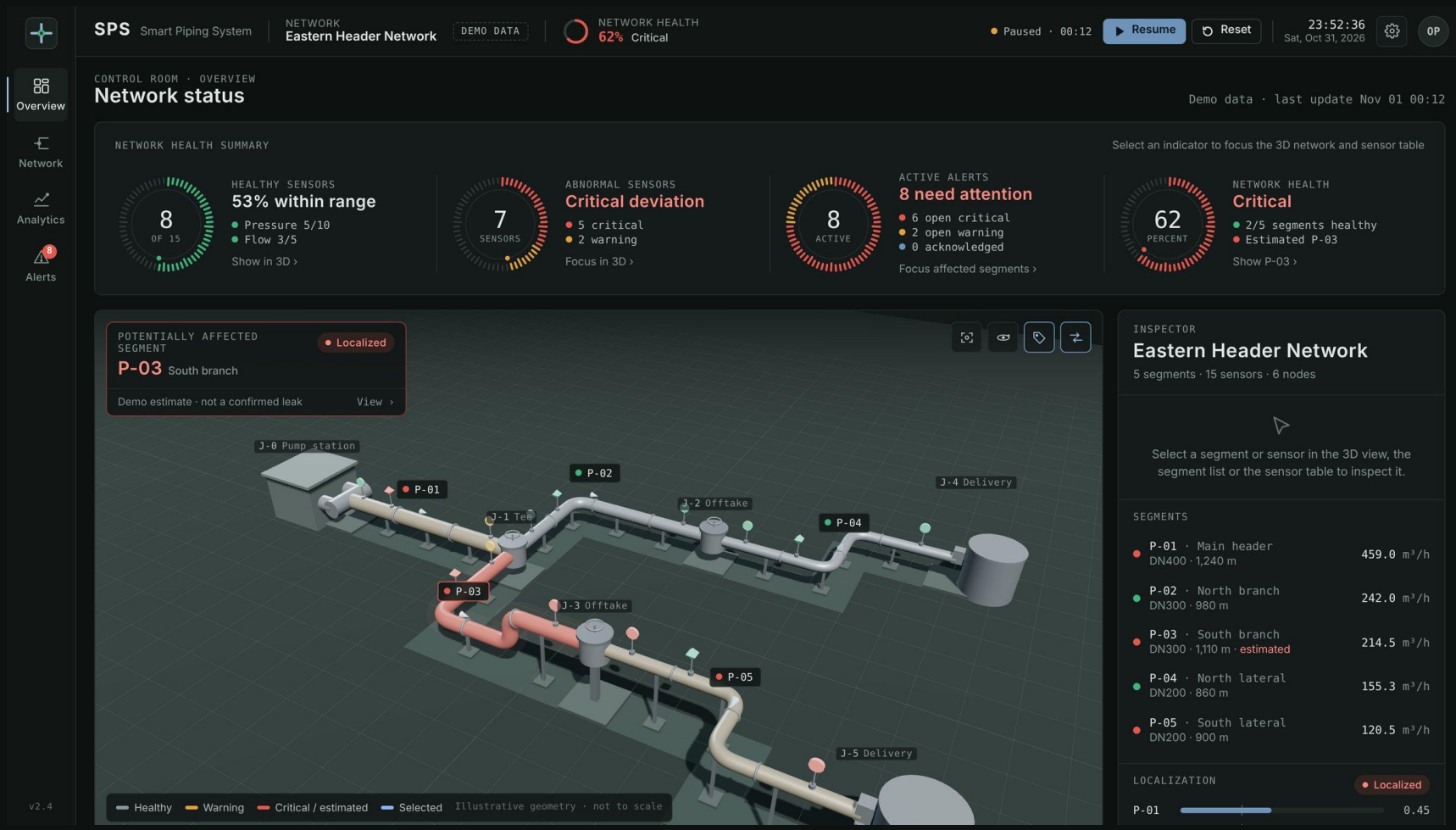
automated tests · 82 pass; 6 need the pinned scikit-learn 1.5.2 or Plotly

Water level

>200 normal · <200 potential leak · =200 boundary; exactly one alert per continuous event (31 tests)

Limit: responses come from an assumed sensor model, not a hydraulic solver or a real network.

One screen: network health first, then the 3D network



Health summary

Healthy and abnormal sensors, active alerts and network health at a glance. Each indicator filters the 3D view.

3D digital twin

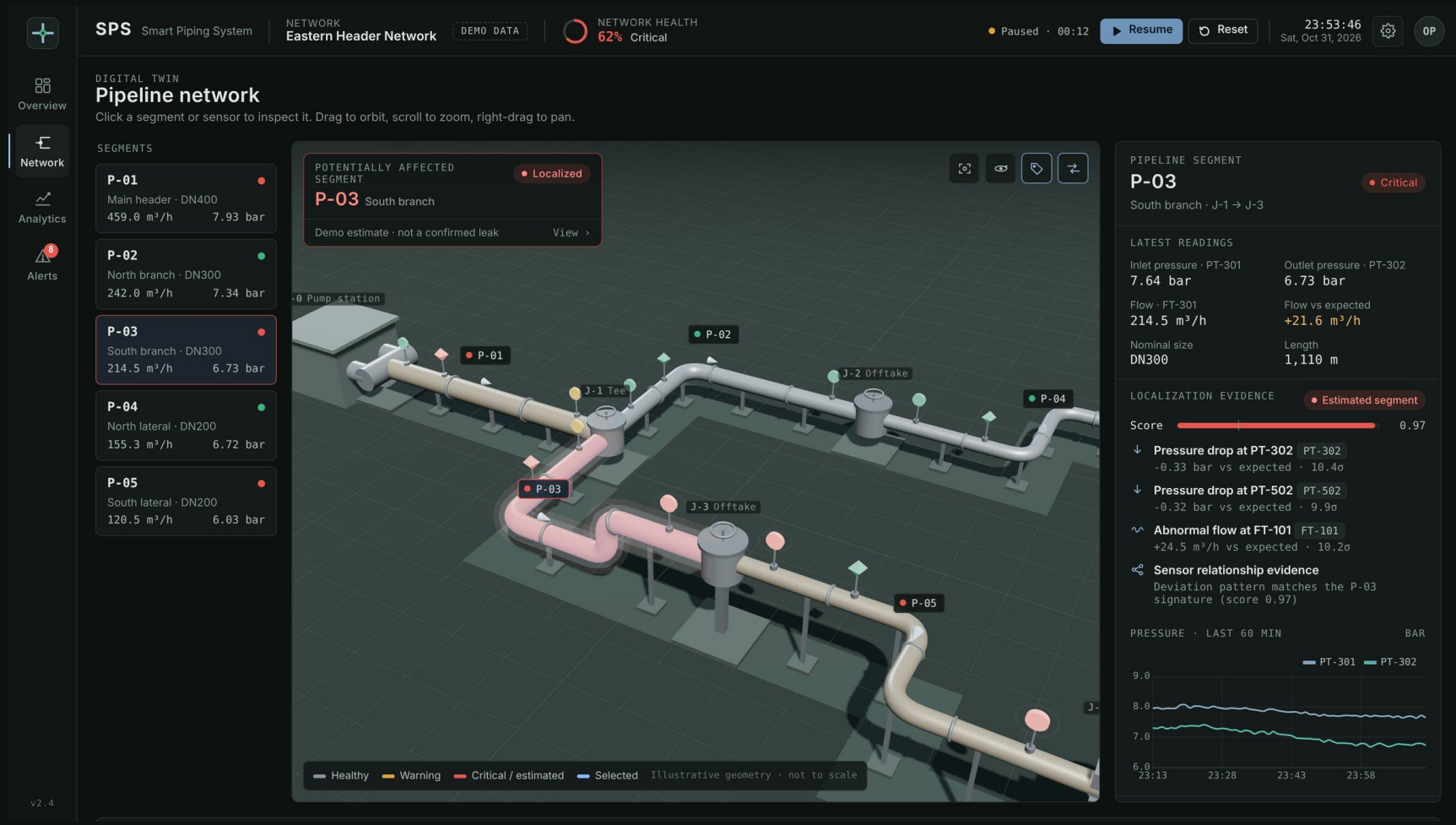
Five segments with sensors; health colours, flow direction and the estimated segment in red.

Inspector

Select any segment or sensor for readings, trends and evidence.

Screens from the SPS web control room · simulated demo data, no physical sensors connected

Click a segment to see why it was estimated



Estimate with a caveat

"Potentially affected segment: P-03", always labelled as a demo estimate, not a confirmed leak.

Evidence, not a black box

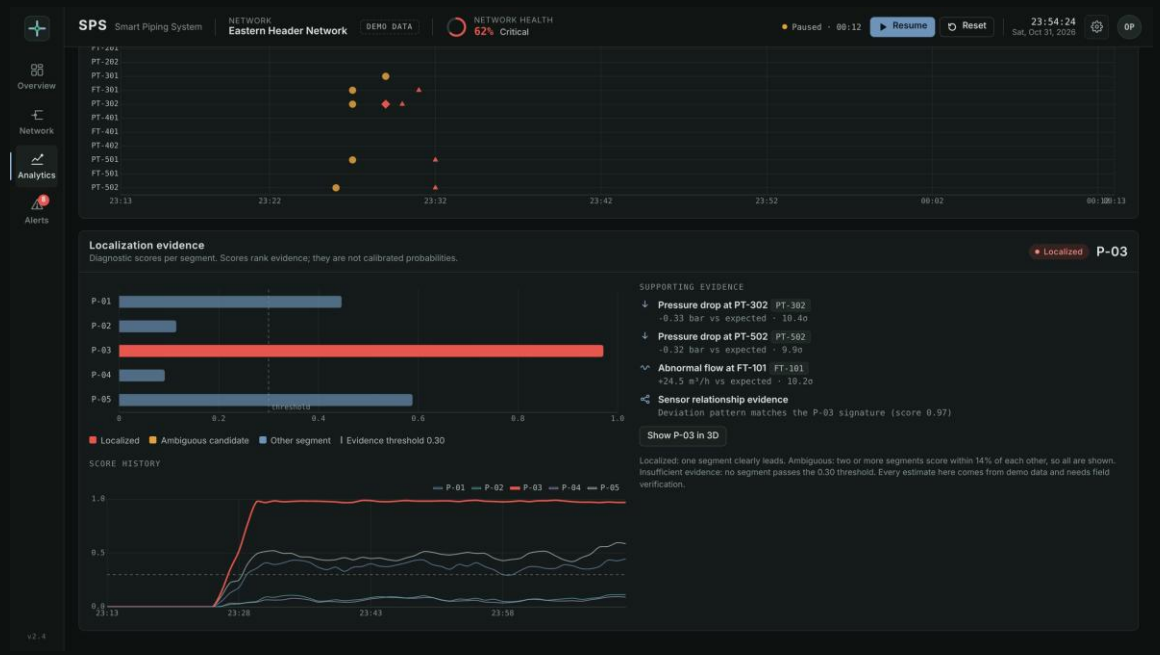
Pressure drops, abnormal flow and the pattern match behind the score.

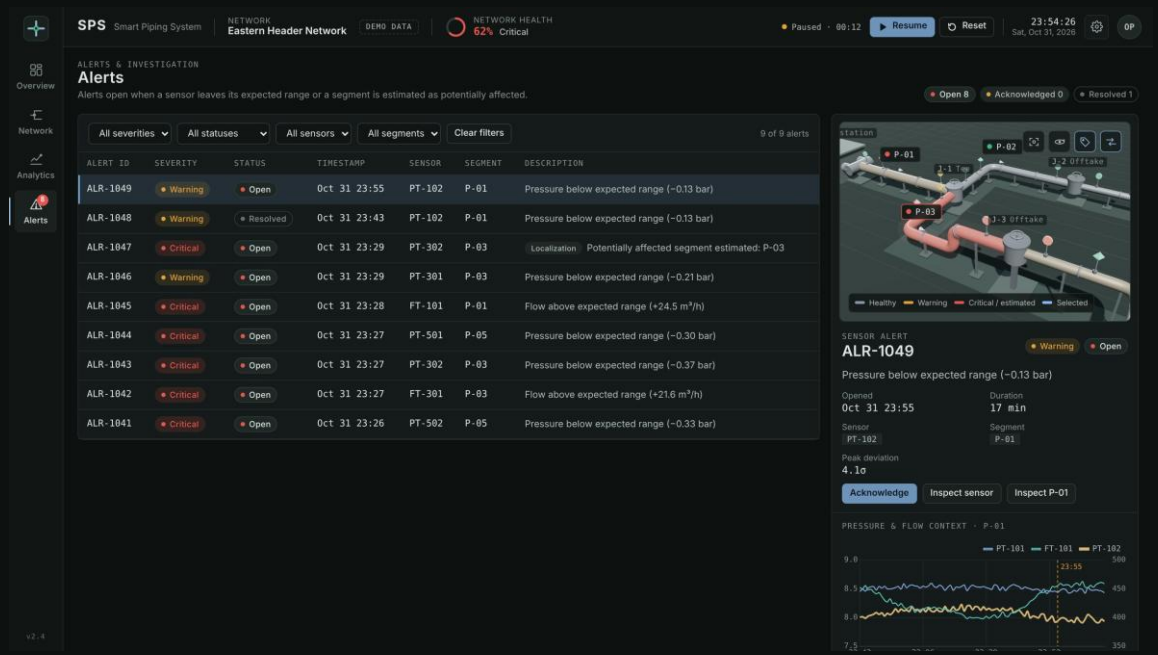
Context

Latest readings, nominal size, length and a 60-minute pressure trend.

Screens from the SPS web control room · simulated demo data, no physical sensors connected

Evidence scores and alerts an operator can act on





Localization evidence

Diagnostic scores per segment against a 0.30 evidence threshold, with score history. Scores rank evidence; they are not probabilities.

Alerts & investigation

Filter by severity, status, sensor and segment; acknowledge alerts; see related readings and the moment each alert opened.

Screens from the SPS web control room · simulated demo data, no physical sensors connected

VALUE PROPOSITION

What SPS can add to an operations team

Earlier attention

Continuous screening of every reading, instead of waiting for someone on site to report a problem.

A smaller search area

A segment estimate with its evidence narrows where crews look first.

Less alarm noise

One alert per continuous event, and boundary readings handled explicitly.

Trustworthy output

Uncertainty is stated: ambiguous and insufficient results are shown as such, never forced.

Target settings, after field validation: water distribution networks, industrial process and cooling-water loops, and large building plumbing.

FUTURE ROADMAP

From software prototype to a field pilot

DONE

Prototype

Backend package, event model baseline, synthetic localization, water-level monitoring, dashboards, 88 tests

NEXT

Hardware rig

Arduino with pressure, flow and HW-038 water-level sensors on a lab pipe loop; live data ingestion

THEN

Own dataset

Controlled leak experiments with timestamps; validation on unseen events

THEN

Better models

Hydraulic network model for localization; calibrated thresholds; drift checks

GOAL

Field pilot

Partner site, SCADA or historian integration, security review, operator feedback

Phases are sequential; dates depend on partners and access to a test rig.

SPS · SMART PIPING SYSTEM

From late discovery to early, explainable indication

- Continuous screening of pressure, flow and water level
- Segment estimates with evidence, and honest uncertainty
- Validated openly; next step is real sensors on a test rig

Thank you

Sources

[1] Liemberger, R. & Wyatt, A. global non-revenue water estimate, reported in "Bursting at the seams: new study highlights the realities of water loss", The Source (IWA), 2018.
https://thesourcemagazine.org/bursting-at-the-seams-new-study-highlights-the-realities-of-water-loss/

[2] Kingdom, B., Liemberger, R. & Marin, P. (2006). The Challenge of Reducing Non-Revenue Water in Developing Countries. World Bank WSS Sector Board Discussion Paper No. 8.
https://documents1.worldbank.org/curated/en/385761468330326484/pdf/394050Reducing1e0water0WSS81PUBLIC1.pdf

[3] National Water Company, Saudi Arabia (2026). NWC Overview, figures through March 2026 (Water Meets Money, Saudi showcase).
https://www.watermeetsmoney.com/wp-content/uploads/2026/05/Day_1_11.15_SaudiShowcase_2NWC.pdf

[4] Shaw, D. et al., Kiefner & Associates (2012). Leak Detection Study, DTPH56-11-D-000001, Final Report 12-173, for U.S. DOT PHMSA.
https://www.phmsa.dot.gov/sites/phmsa.dot.gov/files/docs/technical-resources/pipeline/16691/leak-detection-study.pdf

[5] WDSEventDB: real-time benchmark dataset for event diagnosis in water distribution networks (Zenodo record).
https://zenodo.org/records/17547955

[6] SPS project evaluation reports (reports/*.txt) and test suite (tests/), SPS repository.